

DAY-52-LOG.md — PrepGenie

Day 52 Project Log

Date: Day 52 of 60-Day AI Challenge **Phase:** System Design (no production code written today, by design) **Status:**  Complete

Objective for the Day

Translate the PRD and Executive Summary into concrete, implementation-ready system design artifacts before any code is written — locking in architecture, data model, API contracts, and UI flows as the single source of truth for Days 2-10 of the build phase.

What Was Delivered

1. **ARCHITECTURE.md** — Finalized tech stack with justification, Mermaid component diagram, resume-analysis and mock-interview sequence diagrams, request lifecycle, AI interaction design, external services summary, and key architectural trade-offs.
2. **SCHEMA.md** — Full Postgres schema (4 tables + Supabase Auth), Mermaid ER diagram, RLS policies, Storage bucket convention, and a full traceability matrix validating every PRD user story/requirement against the schema.
3. **API.md** — Complete v1 REST API spec (13 endpoints across auth, resume, interview, dashboard) with request/response shapes, validation rules, auth requirements, and error cases.
4. **UI-WIREFRAMES.md** — User flow diagram, 8-screen inventory with purpose, low-fidelity ASCII wireframes for every screen, navigation structure, and design system notes.
5. **PROJECT-STRUCTURE.md** — Complete folder structure for both `client/` and `server/`, explanation of every major folder, and the reasoning behind the architecture choices.

Key Decisions Made Today

- Confirmed **Express (routes → controllers → services)** layered architecture for the backend — no premature abstraction, no microservices.
- Confirmed **Row-Level Security (RLS)** on every Supabase table as the primary data-isolation mechanism, with app-layer ownership checks as defense-in-depth.
- Confirmed **strict JSON-schema-enforced prompting** as the core AI reliability strategy, with a shared `safeParseAIJson` retry utility used by both the resume analyzer and interview scorer.
- Confirmed all AI-calling endpoints will be rate-limited (10 req/user/hour) to control cost and abuse.
- Validated that **every PRD functional requirement has a corresponding schema element** — no gaps found, no scope changes needed.

Critical Issues Encountered

None. No architecture, schema, or scope changes were required. The PRD and Executive Summary from Day 51 held up completely under detailed system design scrutiny — confirming the original project scoping was sound.

Implementation Blueprint Status

No updates required. The Day 2-10 Implementation Blueprint generated on Day 51 remains fully accurate and aligned with today's system design artifacts. Day 2 (scaffolding, auth, DB schema) can proceed exactly as originally planned, now backed by the detailed SCHEMA.md and ARCHITECTURE.md as reference documents.

What's Next (Day 2 of the Build, per Blueprint)

Begin implementation: project scaffolding, Supabase Auth integration, and database schema creation — using SCHEMA.md's SQL as the exact source for table creation and API.md as the exact contract for the first endpoints.

Git Commit Message

```
docs: add Day 52 system design artifacts (architecture, schema, API, wireframes, structure)
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- Add ARCHITECTURE.md: finalized tech stack, component diagram, data flow sequences (resume analysis + mock interview), request lifecycle, AI interaction design, and external services summary
- Add SCHEMA.md: full Postgres schema (4 tables + RLS policies), ER diagram, storage bucket convention, and PRD requirement traceability matrix
- Add API.md: complete v1 REST API spec covering auth, resume, interview, and dashboard endpoints with request/response/validation/error contracts
- Add UI-WIREFRAMES.md: user flow diagram, 8-screen inventory, low-fidelity wireframes, and navigation structure
- Add PROJECT-STRUCTURE.md: full client/server folder structure with rationale for architectural choices

No production code changes. No scope or architecture deviations from the Day 51 PRD/Executive Summary — all design decisions validated against existing requirements with zero gaps found.

Refs: Day 52/60 — AI Challenge

LinkedIn Post Draft

Day 52/60 of my AI Challenge 🚀

Before writing a single line of production code for my new project, **PrepGenie** (an AI-powered resume analyzer + mock interview coach), I spent today doing something I used to

skip as a student: **proper system design.**

Today's deliverables: 📄 Full architecture doc — tech stack, component diagrams, data flow sequences 🗃️ Complete database schema — with Row-Level Security and a full requirement-traceability matrix 📋 Full REST API specification — 13 endpoints, fully documented before implementation 🎨 Low-fidelity wireframes for every screen + user flow 📁 Production-grade folder structure with reasoning behind every choice

Why this matters: it's tempting to jump straight into coding an AI project. But mapping the architecture, database, and API contracts *before* touching code means the next 8 build days will be faster, cleaner, and far less likely to hit "wait, I need to redesign this" moments.

Next up: Day 2 of the build — project scaffolding, authentication, and standing up the database schema designed today.

Building in public, one day at a time. 💡

#buildinpublic #AI #SystemDesign #BTech #SoftwareEngineering #100DaysOfCode
#PlacementPrep #ClaudeAI
